Supplementary file 1: Full search strategy

Overview for Systematic Review: Systems Approach to Healthcare Design

Database	No. of hits: 10th Aug. 2017	No. of hits: 10th Aug. 2017 – 28th May 2019	No. of hits: 28th May 2019 – 24th Jul. 2020
Medline via OVID	1893	678	487
Embase via OVID	1351	347	295
HMIC via OVID	90	3	2
Health Business Elite via Ebsco	33	n/a	n/a
Web of Science	391	137	113
Scopus	7350	1795	1423
PsycINFO via Ebsco	86	2	3
CINAHL via Ebsco	269	119	72
Total	11,463	3081	2395
Total Deduplicated	8,834	2569	2025

Searches run 10th August 2017, 28th May 2019, 24th July 2020.

Medline

1. (((System or systems or systems-based) adj (approach* or engineering or science or methodolog* or thinking or dynamic* or model* or Whole* or complex* or ergonomics or analys* or theor*)) or (Discrete event simulation) or (sociotechnical or socio-technical)).ti,ab. or exp Systems Analysis/ or exp systems theory/
2. (healthcare or (health adj care) or Medic* or (Health* adj service*) or care or nurs* or (safety adj3 patient*) or treatment outcome* or mortality or morbidity or (Health* adj3 (quality or safety or efficien* or efficac* or performance* or outcome* or deliver* or experience))).ti,ab.
3. (design* or concept* or creat* or plan* or devis* or draft* or propos*).ti,ab.
4. (trial* or longitudinal* or (before adj3 after) or interrupted time series or control* or ((systematic* or literature*) adj review*) or meta-analys* or metaanalys* or (case adj (study* or control*))).mp.

5 1 and 2 and 3 and 4

6 ((201905* or 201906* or 201907* or 201908* or 201909* or 201910* or 201911* or 201912* 2019* or 2020*).dp. or (201905* or 201906* or 201907* or 201908* or 201909* or 201910* or 201911* or 201912* 2019* or 2020*).ez. or (201905* or 201906* or 201907* or 201908* or 201909* or 201910* or 201911* or 201912* or 2020*).ed.)

7 5 and 6

Embase

1. (((System or systems or systems-based) adj (approach* or engineering or science or methodolog* or thinking or dynamic* or model* or Whole* or complex* or ergonomics or analys*

Supplementary file 1: Full search strategy

or theor*)) or (Discrete event simulation) or (sociotechnical or socio-technical)).ti,ab. or exp *Systems Analysis/ or exp *systems theory/
2. (healthcare or (health adj care) or Medic* or (Health* adj service*) or care or nurs* or (safety adj3 patient*) or treatment outcome* or mortality or morbidity or (Health* adj3 (quality or safety or efficien* or efficac* or performance* or outcome* or deliver* or experience))).ti,ab.
3. (design* or concept* or creat* or plan* or devis* or draft* or propos*).ti,ab.
4. (trial* or longitudinal* or (before adj3 after) or interrupted time series or control* or ((systematic* or literature*) adj review*) or meta-analys* or metaanalys* or (case adj (study* or control*))).mp.

5 1 and 2 and 3 and 4

6 ((201905* or 201906* or 201907* or 201908* or 201909* or 201910* or 201911* or 201912* 2019* or 2020*).dc. or (201905* or 201906* or 201907* or 201908* or 201909* or 201910* or 201911* or 201912* or 2020*).dd)

7 5 and 6

Web of Science

TS=((System* N1 approach*) or (system* N1 engineering) or (system* N1 science) or (system* N1 methodology*) or (system* N1 thinking) or (system* N1 dynamic*) or (system* N1 model*) or (system* N1 whole) or (system* N1 complex*) or (system* N1 ergonomics) or (system* N1 analys*) or (system* N1 theor*) or (discrete event simulation) or (sociotechnical) or (socio-technical))

AND

TS=(healthcare or (health care) or Medic* or (Health* service*) or care or nurs* or (safety N3 patient*) or (treatment outcome*) or mortality or morbidity or (Health* N3 quality) or (health* N3 safety) or (Health N3 efficien*) or (health N3 efficac*) or (health N3 performance*) or (health N3 outcome*) or (health N3 deliver*) or (health N3 experience))

AND

TS=(design* or concept* or creat* or plan* or devis* or draft* or propos*)

AND

TS=(trial* or longitudinal* or (before N3 after) or (interrupted time series) or control* or (systematic* review*) or (literature review*) or meta-analys* or metaanalys* or (case study*) or (case control*))

AND

#4 AND #3 AND #2 AND #1
Refined by: PUBLICATION YEARS: (2019 OR 2020)

CINAHL

Supplementary file 1: Full search strategy

#	Query
S7	S3 AND S4 AND S5 AND S6 Limiters - Published Date: 20190501-20201231
S6	trial* or longitudinal* or (before N3 after) or (interrupted time series) or control* or (systematic* review) or (literature review*) or meta-analys* or metaanalys* or (case study*) or (case control*)
S5	TI (design* or concept* or creat* or plan* or devis* or draft* or propos*) OR AB (design* or concept* or creat* or plan* or devis* or draft* or propos*)
S4	TI ((healthcare or (health adj care) or Medic* or (Health* adj service*) or care or nurs* or (safety N3 patient*) or treatment outcome* or mortality or morbidity or (Health* N3 (quality or safety or efficien* or efficac* or performance* or outcome* or deliver* or experience)))) OR AB ((healthcare or (health adj care) or Medic* or (Health* adj service*) or care or nurs* or (safety N3 patient*) or treatment outcome* or mortality or morbidity or (Health* N3 (quality or safety or efficien* or efficac* or performance* or outcome* or deliver* or experience))))
S3	S1 OR S2
S2	TI (((System or systems or systems-based) N (approach* or engineering or science or methodolog* or thinking or dynamic* or model* or Whole* or complex* or ergonomics or analys* or theor*)) or (Discrete event simulation) or (sociotechnical or socio-technical)) OR AU (((System or systems or systems-based) N (approach* or engineering or science or methodolog* or thinking or dynamic* or model* or Whole* or complex* or ergonomics or analys* or theor*)) or (Discrete event simulation) or (sociotechnical or socio-technical))
S1	(MH "Systems Theory+") OR (MH "Systems Analysis+")

PsycINFO

#	Query
	Limiters - Published Date: 20190501-20201231
S7	S3 AND S4 AND S5 AND S6
S6	trial* or longitudinal* or (before N3 after) or (interrupted time series) or control* or (systematic* review) or (literature review*) or meta-analys* or metaanalys* or (case study*) or (case control*)

Supplementary file 1: Full search strategy

S5	TI (design* or concept* or creat* or plan* or devis* or draft* or propos*) OR AB (design* or concept* or creat* or plan* or devis* or draft* or propos*)
S4	TI ((healthcare or (health adj care) or Medic* or (Health* adj service*) or care or nurs* or (safety N3 patient*) or treatment outcome* or mortality or morbidity or (Health* N3 (quality or safety or efficien* or efficac* or performance* or outcome* or deliver* or experience)))) OR AB ((healthcare or (health adj care) or Medic* or (Health* adj service*) or care or nurs* or (safety N3 patient*) or treatment outcome* or mortality or morbidity or (Health* N3 (quality or safety or efficien* or efficac* or performance* or outcome* or deliver* or experience))))
S3	S1 OR S2
S2	TI (((System or systems or systems-based) N (approach* or engineering or science or methodolog* or thinking or dynamic* or model* or Whole* or complex* or ergonomics or analys* or theor*)) or (Discrete event simulation) or (sociotechnical or socio-technical)) OR AU (((System or systems or systems-based) N (approach* or engineering or science or methodolog* or thinking or dynamic* or model* or Whole* or complex* or ergonomics or analys* or theor*)) or (Discrete event simulation) or (sociotechnical or socio-technical))
S1	(DE "Systems Theory" OR DE "Systems Neuroscience") OR (DE "Systems Analysis")

HMIC – Health Management Information Consortium

1. (((System or systems or systems-based) adj (approach* or engineering or science or methodolog* or thinking or dynamic* or model* or Whole* or complex* or ergonomics or analys* or theor*)) or (Discrete event simulation) or (sociotechnical or socio-technical)).ti,ab. or exp Systems Analysis/ or exp systems theory/
2. (healthcare or (health adj care) or Medic* or (Health* adj service*) or care or nurs* or (safety adj3 patient*) or treatment outcome* or mortality or morbidity or (Health* adj3 (quality or safety or efficien* or efficac* or performance* or outcome* or deliver* or experience))).ti,ab.
3. (design* or concept* or creat* or plan* or devis* or draft* or propos*).ti,ab.
4. (trial* or longitudinal* or (before adj3 after) or interrupted time series or control* or ((systematic* or literature*) adj review*) or meta-analys* or metaanalys* or (case adj (study* or control*))).mp.

5 1 and 2 and 3 and 4

6 ((201905* or 201906* or 201907* or 201908* or 201909* or 201910* or 201911* or 201912* 2019* or 2020*).yr. or (201905* or 201906* or 201907* or 201908* or 201909* or 201910* or 201911* or 201912* or 2020*).dp.)

7 5 and 6

Health Business Elite this database is no longer available

#	Query
S7	S3 AND S4 AND S5 AND S6

Supplementary file 1: Full search strategy

S6	trial* or longitudinal* or (before N3 after) or (interrupted time series) or control* or (systematic* review) or (literature review*) or meta-analys* or metaanalys* or (case study*) or (case control*)
S5	TI (design* or concept* or creat* or plan* or devis* or draft* or propos*) OR AB (design* or concept* or creat* or plan* or devis* or draft* or propos*)
S4	TI ((healthcare or (health adj care) or Medic* or (Health* adj service*) or care or nurs* or (safety N3 patient*) or treatment outcome* or mortality or morbidity or (Health* N3 (quality or safety or efficien* or efficac* or performance* or outcome* or deliver* or experience)))) OR AB ((healthcare or (health adj care) or Medic* or (Health* adj service*) or care or nurs* or (safety N3 patient*) or treatment outcome* or mortality or morbidity or (Health* N3 (quality or safety or efficien* or efficac* or performance* or outcome* or deliver* or experience))))
S3	S1 OR S2
S2	TI (((System or systems or systems-based) N (approach* or engineering or science or methodolog* or thinking or dynamic* or model* or Whole* or complex* or ergonomics or analys* or theor*)) or (Discrete event simulation) or (sociotechnical or socio-technical)) OR AU (((System or systems or systems-based) N (approach* or engineering or science or methodolog* or thinking or dynamic* or model* or Whole* or complex* or ergonomics or analys* or theor*)) or (Discrete event simulation) or (sociotechnical or socio-technical))
S1	(DE "SYSTEMS theory" OR DE "ASYMPTOTIC theory of system theory" OR DE "AUTOPOIESIS" OR DE "BIOLOGICAL systems" OR DE "CHAOS theory" OR DE "CONSTRUCTAL theory" OR DE "CYBERNETICS" OR DE "CYBERNETICS -- Social aspects" OR DE "DISCRETE systems" OR DE "EARTH system science" OR DE "LINEAR systems" OR DE "NONLINEAR systems" OR DE "OPEN systems theory" OR DE "OPERATIONS research" OR DE "SOCIAL dynamics" OR DE "SOCIAL systems" OR DE "SYNERGETICS" OR DE "SYSTEM analysis" OR DE "SYSTEMS engineering") OR (DE "SYSTEM analysis" OR DE "BEHAVIORAL systems analysis" OR DE "BOND graphs" OR DE "BUSINESS requirements analysis" OR DE "CONTROL theory (Mathematics)" OR DE "COUPLED problems (Complex systems)" OR DE "DECOMPOSITION method (Mathematics)" OR DE "DISCRETE-time systems" OR DE "DISTRIBUTED parameter systems" OR DE "ELECTRIC networks" OR DE "FLOW charts" OR DE "FLOWGRAPHS" OR DE "FUZZY systems" OR DE "HIPO technique" OR DE "INFORMATION modeling" OR DE "LARGE scale systems" OR DE "MAPS design technology" OR DE "MEMORYLESS systems" OR DE "SADT (System analysis)" OR DE "SAMPLE path analysis" OR DE "SMALL scale system" OR DE "SPATIAL systems" OR DE "STATE-space methods" OR DE "STOCHASTIC systems" OR DE "STRUCTURED techniques of electronic data processing" OR DE "SUBJECTIVE transfer function method" OR DE "SUPERPOSITION principle (Physics)" OR DE "SWITCHING theory" OR DE "SYSTEM identification" OR DE "SYSTEMS design" OR DE "TIME-domain analysis" OR DE "TRANSMISSION network calculations" OR DE "UNCERTAIN systems")

Scopus

Supplementary file 1: Full search strategy

(TITLE-ABS ((system* W/1 approach*) OR (system* W/1 engineering) OR (system* W/1 science) OR (system* W/1 methodology*) OR (system* W/1 thinking) OR (system* W/1 dynamic*) OR (system* W/1 model*) OR (system* W/1 whole) OR (system* W/1 complex*) OR (system* W/1 ergonomics) OR (system* W/1 analys*) OR (system* W/1 theor*) OR ("discrete event simulation") OR (sociotechnical) OR (socio-technical))) AND (TITLE-ABS (design* OR concept* OR creat* OR plan* OR devis* OR draft* OR propos*)) AND (TITLE-ABS (trial* OR longitudinal* OR (before W/3 after) OR ("interrupted Time series") OR control* OR ("systematic* review*") OR ("literature review*") OR meta-analys* OR metaanalys* OR ("case study*") OR ("case control*")))) AND (TITLE-ABS (healthcare OR ("health care") OR medic* OR ("health* service*") OR care OR nurs* OR (safety W/3 patient*) OR ("treatment outcome*") OR mortality OR morbidity OR (health* W/1 (quality OR safety OR efficien* OR efficac* OR performance* OR outcome* OR deliver* OR experience))))

Limit to 2019, 2020